LESSON PLAN

BASIC RADIO & AVIONICS (25)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Basic requirement of instrument in a typical aircraft and types
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Basic requirement of instrument in a typical aircraft and types
5. Specific Objectives	25.1.1.1 Trainee will be able to understand basic instrument gauges, instrument type auto pilot and maintenance.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of instrument types from IB; Instructor supervision required for safety
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an aircraft instrument?
(b) Name one essential instrument found in a typical cockpit.
(c) Why are instruments required in a typical aircraft?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic requirements of aircraft instruments • Instrument types and classifications • Instrument gauges and displays • Auto pilot system basics • Instrument maintenance overview	L & Ds	PPT, IB	15 min
(b)	• Detailed analysis of instrument gauges • Operational principles of auto pilot • Maintenance procedures for instruments • Troubleshooting common instrument faults • Safety checks for instrument systems	L & Ds	PPT, PDF	20 min
(c)	• Summary of instrument requirements • Review of instrument types • Key maintenance points • Auto pilot integration basics • Final Q&A session	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the basic requirement of an instrument in a typical aircraft.
(b) Differentiate between the types of instruments discussed.

12. Home Assignment and Follow Up	Prepare a brief note on the purpose and principle operation of aircraft pressure gauges, pitot-static system, altimeter, ASI, and VSI for the next lesson.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Purpose and principle operation of aircraft pressure gauges, pilot static system, altimeter, ASI, and VSI. Practical on Pitot & Static instruments. Safety precautions while working on system	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min